

Day 2 Of Communication System Class

Performance Parameters of Communication System

1. Noise

- Definition: Unwanted electrical signal that interferes with the original message.
- Measurement: Signal-to-Noise Ratio (SNR).
- Note: Noise is always present; the goal is to maximize SNR.

2. Distortion

- Definition: Unwanted change in the shape or characteristics of the transmitted signal.
- Effect: Received signal differs from the transmitted signal.
- Mark in notes: Often marked with X or (φ) .

3. Delay (Latency)

- Components:
- Processing Delay: Time taken by devices to process the signal/packet.
- Propagation Delay: Time for the signal to travel through the medium (depends on distance and medium speed).
- Jitter: Variation in packet delay (inconsistent delays).
- Overall: Called Latency.

4. Data Rate / Baud Rate / Throughput

- Data Rate: Number of bits transmitted per second (bps).
- Baud Rate: Number of signal changes (symbols) per second.
- Throughput: Actual successful/ effective data transmission rate (usually $\leq$ data rate due to overhead, errors, etc.).

5. Bandwidth (BW)

- **Analog Signal:**

$$BW = f_{\max} - f_{\min} \quad (\text{in Hz})$$

- **Digital Signal:** Depends on sampling.

Nyquist Sampling Theorem:

$$f_s = 2 \times f_{\max}$$

- **PCM Bit Rate:**

$$\text{Bit Rate} = n \times f_s = n \times 2 \times f_{\max}$$

where n = number of bits per sample.

Examples:

- **FM Radio:** 88 MHz – 108 MHz (BW = 20 MHz).
- **Voice Signal:** 300 Hz – 3400 Hz $\approx$ **4 kHz** (with guard band consideration).
- **Voice PCM:**
 $8 \times 2 \times 4000 = 64,000 \text{ bps} = 64 \text{ kbps}$ (standard telephone quality).

6. Channel Capacity (CC) – Shannon-Hartley Theorem

According to **Claude Shannon**:

$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

Where:

- C = Channel Capacity (bps)
- B = Bandwidth (Hz)
- $\frac{S}{N}$ = Signal-to-Noise Ratio (linear, not dB)
- Assumes **Gaussian noise**.

Note: S = Signal power, N = Noise power.

7. Signal-to-Noise Ratio (SNR)

- In dB:

$$\text{SNR (dB)} = 10 \log_{10} \left(\frac{S}{N} \right)$$

- Interpretation:

- Positive (+): Signal > Noise (good).
- Negative (-): Signal < Noise (poor).

- Linear form: $\frac{S}{N}$.

[Most Probable Question]

1. Calculate the theoretical channel capacity of a communication link if its bandwidth is 4000 Hz and its operating SNR is 20 dB.

Solution:

1. Convert SNR to linear:

$$\frac{S}{N} = 10^{20/10} = 10^2 = 100$$

2. Apply Shannon formula:

$$C = 4000 \times \log_2(1 + 100) = 4000 \times \log_2(101)$$

3. $\log_2(101) \approx 6.6582$

$$C \approx 4000 \times 6.6582 = 26,632.8 \text{ bps} \approx 26.63 \text{ kbps}$$

Final Answer:

$C \approx 26.63 \text{ kbps}$
